

Trust, Decision Making, and Black Box Algorithms in Medicine

Introduction

Artificial intelligence is now part of many areas of medicine. Computers can help doctors predict illnesses, sort through data, and suggest treatments. One key concern is the use of "black box" algorithms. These are computer programs whose internal workings are so complex that even experts cannot easily explain how they reach their conclusions. Many of these algorithms, like deep neural networks, are powerful, as they can diagnose diseases from images or provide treatment recommendations. But the fact that their logic is hidden raises difficult questions for doctors, patients, and hospitals.

This reflection will consider those issues using ideas from *Duran and Jongsma, 2021*, who argue for the importance of "computational reliabilism", and will apply them to a realistic medical scenario. The discussion focuses on how doctors can trust these systems, whether they can act on their advice, and what this means for responsibility and patient choice.

Scenario Description

Imagine a doctor named Dr. Lee working in the emergency department. A patient, Mr. Brown arrives with chest pain. Dr. Lee uses a new machine learning tool designed to help detect heart attacks. After entering Mr. Brown's test results and symptoms, the AI system gives a strong indication for a heart attack and recommends immediate treatment with blood thinners. Dr. Lee, looking at the same test results and ECG herself, does not see the typical signs. She faces a dilemma, the AI system has performed very well in tests, showing better accuracy than most doctors, but she does not know what factors in Mr. Brown's data led to the algorithm's decision. When Mr. Brown's family asks for an explanation, Dr. Lee can only say that the system has been proven reliable in similar cases; she cannot break down its reasoning.

Here, Dr. Lee must decide what to do: should she trust and follow the AI's recommendation? If she does, can she justify that decision ethically and professionally to her patient?

Ethical Analysis

This problem is addressed by discussing two related ideas: epistemic trust (can we believe the result?) and moral justification (should we base actions on the result?). (*Duran and Jongsma, 2021*) They argue that even if an AI algorithm is a black box, doctors can be justified in trusting its outputs. Not because they understand how it works, but because the system's results have been shown, through validation, expert review, and evidence, to be accurate most of the time. This is called "computational reliabilism". In other words, if a tool

has been tested in many situations, has stood up to review, and performs reliably, then it is not unreasonable for doctors to take its advice seriously. Even if the inner logic is hidden.

However, trust in the result is not the same as having a solid ethical ground for acting on it. (*Duran and Jongsma, 2021*) In medicine, doing the right thing involves more than just statistical evidence. Doctors are expected to make decisions that consider the specific situation, the possible risks, and, importantly, the patient's own wishes and values. Even when an AI system gives a clear recommendation, the doctor should use her own judgement, weigh the risks and benefits, and talk with the patient about different options.

In the example of Dr. Lee and Mr. Brown, the main ethical problems are responsibility and consent. If Dr. Lee follows the AI's recommendation and something goes wrong, such as an unnecessary treatment causing harm, she remains responsible. She cannot simply blame the machine, especially if she cannot explain to her patient or other healthcare staff what led to that decision. On the other hand, if she ignores the AI and the patient does have a heart attack, she could be blamed for neglecting a tool known to be highly accurate. The challenge is made harder by the fact that AI systems cannot provide human-understandable reasons for their advice.

Another concern relates to the patient's autonomy. Patients have the right to understand the reasons for their treatment and to take part in these decisions. If an answer from an AI system cannot be explained in normal language, it may become very hard for the patient to make an informed choice. This problem does not come only from the opaqueness of the algorithm, but also from how results are used in practice. (*Duran and Jongsma, 2021*) For example, if the doctor's discussion with the patient considers the possible treatments, describing both the reasons and risks for each, this respects patient autonomy even if the AI's logic is not fully clear. The main danger is if the doctor uses a black box system's result as the only option and does not engage with the patient about alternatives.

Relying too much on these tools could lead to a shift in medicine where doctors feel less responsible for their decisions or even allow AI to make final choices. (*Duran and Jongsma, 2021*) The authors warn that this is undesirable, as human judgement, the ability to understand context, and ethical consideration should not be replaced by AI, even a highly reliable one.

Critical Position and Reflection

For Dr. Lee, recognising the reliability of the AI system is important, but so is her professional role. The best course of action would be to use the AI result as a strong input to be considered, but not as a final decision. She should weigh the AI's result against her own knowledge, consult guidelines, and, if possible, talk through the options with Mr. Brown and his family, explaining both the uncertainty and the reasons for considering immediate treatment. She should say that while the test results are not entirely clear to her, the hospital's AI tool has proven very accurate in cases like this, and that this is the key reason for recommending the stronger treatment.

In this way, trust in the black box AI is possible when the tool is proven reliable, but medical decisions must still include human judgement, patient input, and open discussion. Transparency in terms of process, such as information about how the AI is tested or how often it gets things right, can help replace some of the lack of transparency in terms of logic. As the use of complex AI increases in medicine, it becomes even more essential for doctors to understand their strengths, limitations, and the need for responsible, patient-focused discussion.

Conclusion

The use of black box algorithms in healthcare can be justified if they are shown to be reliable, but doctors remain responsible for their actions and for explaining and discussing treatment options with patients. Acting on AI advice should be done with caution, reflection, and clear communication; not just blind obedience. The promise of AI should not come with the cost of patient understanding or professional accountability, but can be a valuable support to both.

References

Durán, J. M., & Jongsma, K. R. (2021). Who is afraid of black box algorithms? On the epistemological and ethical basis of trust in medical AI. *Journal of Medical Ethics*, 47(5), medethics-2020-106820. <https://doi.org/10.1136/medethics-2020-106820>